Rachel Hautz

Lecture 19 3/4/26

Complex Exponential Solution

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t} \quad \Rightarrow \quad C_1 = \tfrac{1}{2}(B_1 - iB_2) \qquad C_2 = \tfrac{1}{2}(B_1 + iB_2) \quad , \quad B_{1,2} \in \mathbb{R}$$

$$x(t) = B_1 \cos\omega t + B_2 \sin\omega t$$

$$\Rightarrow \quad C_2 = C_1^*$$

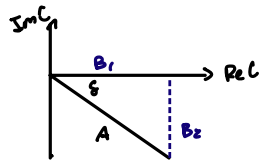
$$\Rightarrow \quad x(t) = \underbrace{C_1 e^{i\omega t}}_{z} + \underbrace{C_1^* e^{-i\omega t}}_{z^*}$$

For any $z \in \mathbb{C}$, $z + z^* = x + iy + x - iy = 2x = 2\mathrm{Re}(z)$

$$x(t) = z + z^* = 2\mathrm{Re}(z) = 2\mathrm{Re}\left[C_1 e^{i\omega t}\right] \qquad \text{let } C \equiv 2C_1$$

$$x(t) = \mathrm{Re}\left[C e^{i\omega t}\right] \qquad C = B_1 - iB_2$$

$$x(t) = \mathrm{Re}\left[A e^{i(\omega t - \delta)}\right] = A\cos(\omega t - \delta)$$



$$\Rightarrow \quad C = Ae^{-i\delta}$$

Total Energy:

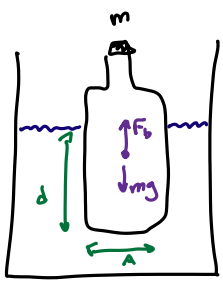
$$E = T + U = \tfrac{1}{2} m \dot{x}^2 + \tfrac{1}{2} k x^2 = \tfrac{1}{2} m \left[-\omega A \sin(\omega t - \delta)\right]^2 + \tfrac{1}{2} k A^2 \cos^2(\omega t - \delta)$$

$$= \tfrac{1}{2} k A^2 \sin^2(\omega t - \delta) + \tfrac{1}{2} k A^2 \cos^2(\omega t - \delta)$$

$$= \tfrac{1}{2} k A^2 \underbrace{\left[\sin^2(\omega t - \delta) + \cos^2(\omega t - \delta)\right]}_{=1}$$

$$= \tfrac{1}{2} k A^2 = \text{constant} \quad (\text{as it should be})$$

Example:



Released d below surface, where d_0 = position of equilibrium. What is τ?

$$F = -mg + \rho A g d \qquad \text{at } d = d_0,\ F = 0 \Rightarrow mg = \rho A g d_0$$

ρ ↑ density of water

$$\frac{m}{\rho A} = d_0$$

let $d = d_0 + x$

↑ displacement from equilibrium

$$\Rightarrow m\ddot{x} = mg - \rho g A (d_0 + x)$$

$$= \cancel{mg} - \cancel{\rho g A d_0} - \rho g A x = -\rho g A x$$

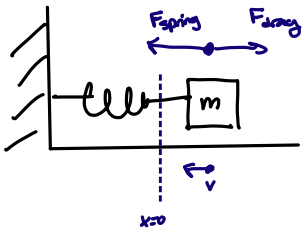
$$\ddot{x} = -\frac{\rho g A}{m} x = -\frac{g}{d_0} x$$

$\omega = \sqrt{g/d_0}$ and we have SHM

$$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{d_0}{g}}$$

Damped Oscillations

- Consider a SHO subject to drag force



$$\vec{F}_{drag} = -b\vec{v}$$
$$\vec{F}_{spring} = -k\vec{x}$$

$$m\vec{a} = \vec{F}_{spring} + \vec{F}_{drag}$$
$$m\ddot{x} = -kx - b\dot{x}$$

To simplify our algebra,
let $2\beta \equiv b/m$
$\omega_0 \equiv \sqrt{k/m}$

$$\Rightarrow \underbrace{m\ddot{x}}_{ma} + \underbrace{b\dot{x}}_{drag} + \underbrace{kx}_{spring} = 0 \Rightarrow \ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$$

$$\Rightarrow \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$$ } EOM for damped oscillator

How do we solve this dif eq? $\Rightarrow$ linear, homogeneous, 2nd order

Strategy: guess two solutions $\Rightarrow$ the general solution is a linear combination!

Guess: $x(t) = e^{rt}$ $\Rightarrow$ Plug this into EOM to find r

EOM: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$ Need $\dot{x} = re^{rt}$
$\ddot{x} = r^2 e^{rt}$

$$\Rightarrow (r^2 e^{rt}) + 2\beta(re^{rt}) + \omega_0^2(e^{rt}) = 0$$
$$r^2\cancel{e^{rt}} + 2\beta r\cancel{e^{rt}} + \omega_0^2\cancel{e^{rt}} = 0$$

$\Rightarrow r^2 + 2\beta r + \omega_0^2 = 0$ $\Rightarrow$ tells us r! Sometimes this is called the auxiliary equation

$$\Rightarrow r = \frac{-2\beta \pm \sqrt{(2\beta)^2 - 4\omega_0^2}}{2}$$

$$r = -\beta \pm \sqrt{\beta^2 - \omega_0^2} = \begin{cases} -\beta + \sqrt{\beta^2-\omega_0^2} \equiv r_1 \\ -\beta - \sqrt{\beta^2-\omega_0^2} \equiv r_2 \end{cases}$$ } two values of $r \Rightarrow$ two solutions!

$$x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

$$x(t) = C_1 e^{(-\beta + \sqrt{\beta^2-\omega_0^2})t} + C_2 e^{(-\beta - \sqrt{\beta^2-\omega_0^2})t} = C_1 e^{-\beta t + \sqrt{\beta^2-\omega_0^2}\,t} + C_2 e^{-\beta t - \sqrt{\beta^2-\omega_0^2}\,t}$$

$$x(t) = C_1 e^{-\beta t} e^{\sqrt{\beta^2-\omega_0^2}\,t} + C_2 e^{-\beta t} e^{-\sqrt{\beta^2-\omega_0^2}\,t}$$

$$x(t) = e^{-\beta t}\left[C_1 e^{\sqrt{\beta^2-\omega_0^2}\,t} + C_2 e^{-\sqrt{\beta^2-\omega_0^2}\,t}\right]$$ $\Rightarrow$ What does this look like?

Cases: (I) $\beta = 0$, (II) $\beta^2 - \omega_0^2 < 0$, (III) $\beta^2 - \omega_0^2 > 0$, (IV) $\beta^2 - \omega_0^2 = 0$

↑ simplest case first

I: Undamped Oscillation, $\beta = 0$

$$x(t) = e^0 \left[C_1 e^{\sqrt{-\omega_0^2}\, t} + C_2 e^{-\sqrt{-\omega_0^2}\, t} \right]$$

$$x(t) = C_1 e^{i\omega_0 t} + C_2 e^{-i\omega_0 t}$$ ⇒ simple harmonic oscillator soln from before (just a mass on a spring)

This make sense. Remember

$2\beta = \frac{b}{m}$ ⇒ $\beta = 0$ gives $b = 0$, $\vec{F}_{drag} = b\vec{v}$

⇒ $\vec{F}_{drag} = 0$ ⇒ undamped oscillation

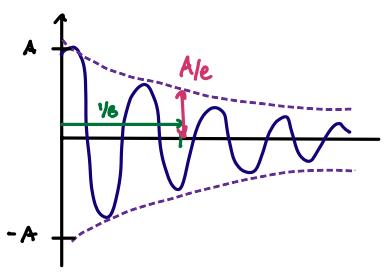
II. Weak Damping (under damping

$\sqrt{\beta^2 - \omega_0^2} \notin \mathbb{R}$ or $\beta^2 - \omega_0^2 < 0$ or $\boxed{\beta < \omega_0}$

$$\Rightarrow \sqrt{\underbrace{\beta^2 - \omega_0^2}_{\text{negative}}} = i\sqrt{\underbrace{\omega_0^2 - \beta^2}_{\text{positive}}}$$ Define $\omega_1 \equiv \sqrt{\omega_0^2 - \beta^2}$

$$x(t) = e^{-\beta t} \underbrace{\left(C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t} \right)}_{\text{can rewrite: } A\cos(\omega_1 t - \delta)}$$

$$x(t) = A e^{-\beta t} \cos(\omega_1 t - \delta)$$



Envelope given by $A(t) = Ae^{-\beta t}$ $A(0) = A$

when $t = 1/\beta$, $A(t) = Ae^{-1} = A\left(\frac{1}{e}\right) = \frac{1}{e}A = \frac{1}{e}A(0)$

$A(1/\beta) = \frac{1}{e}A(0)$

⇒ $\beta \equiv$ decay parameter

III. Strong Damping (over damping)

$\sqrt{\beta^2 - \omega_0^2} \in \mathbb{R}$ or $\beta^2 - \omega_0^2 > 0$ or $\boxed{\beta > \omega_0}$

$$x(t) = e^{-\beta t} \left(C_1 e^{\sqrt{\beta^2 - \omega_0^2}\, t} + C_2 e^{-\sqrt{\beta^2 - \omega_0^2}\, t} \right)$$

$$x(t) = C_1 e^{-(\beta - \sqrt{\beta^2 - \omega_0^2})t} + C_2 e^{-(\beta + \sqrt{\beta^2 - \omega_0^2})t}$$

Argument of exponent is real ⇒ no oscillations!